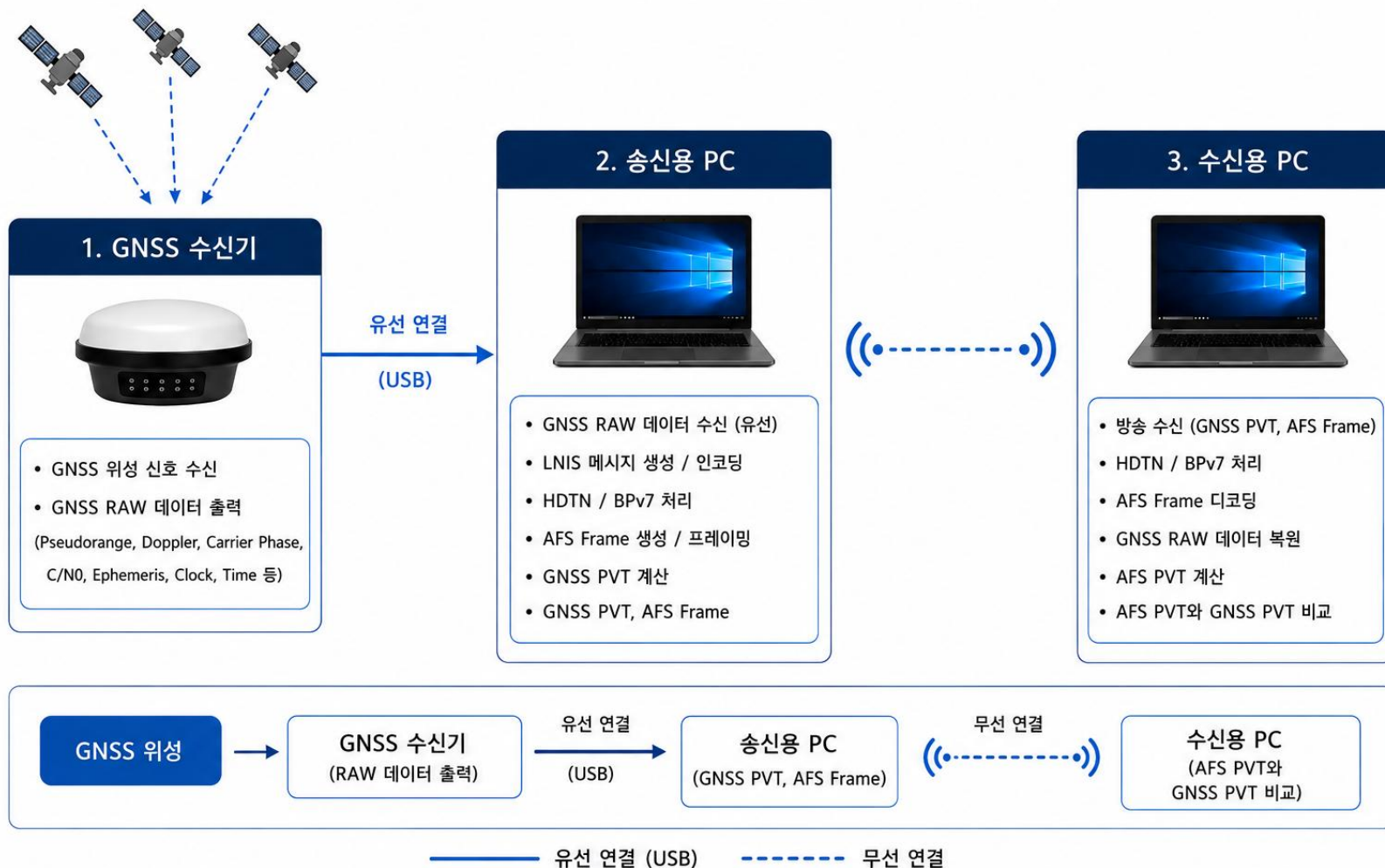
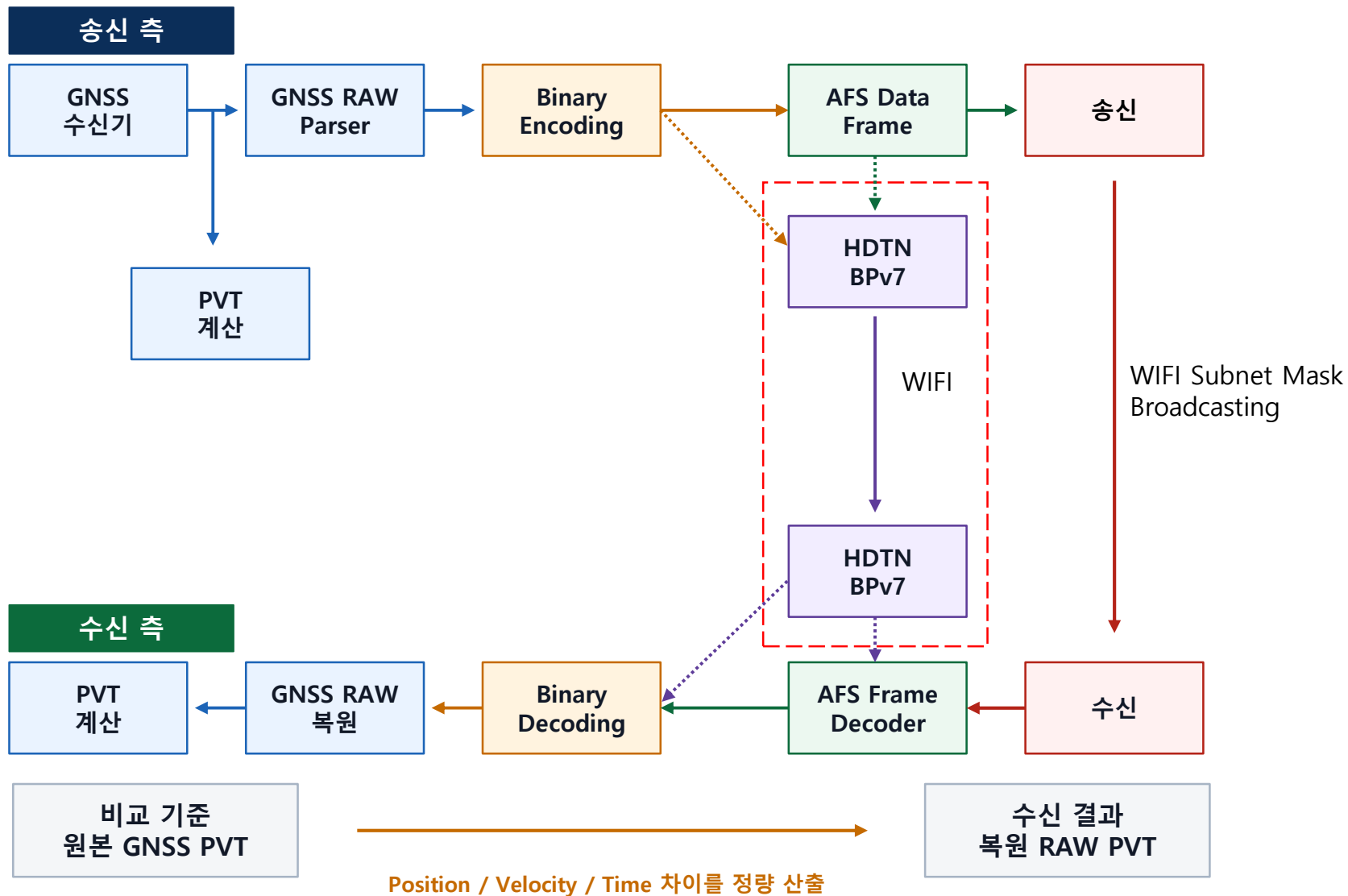


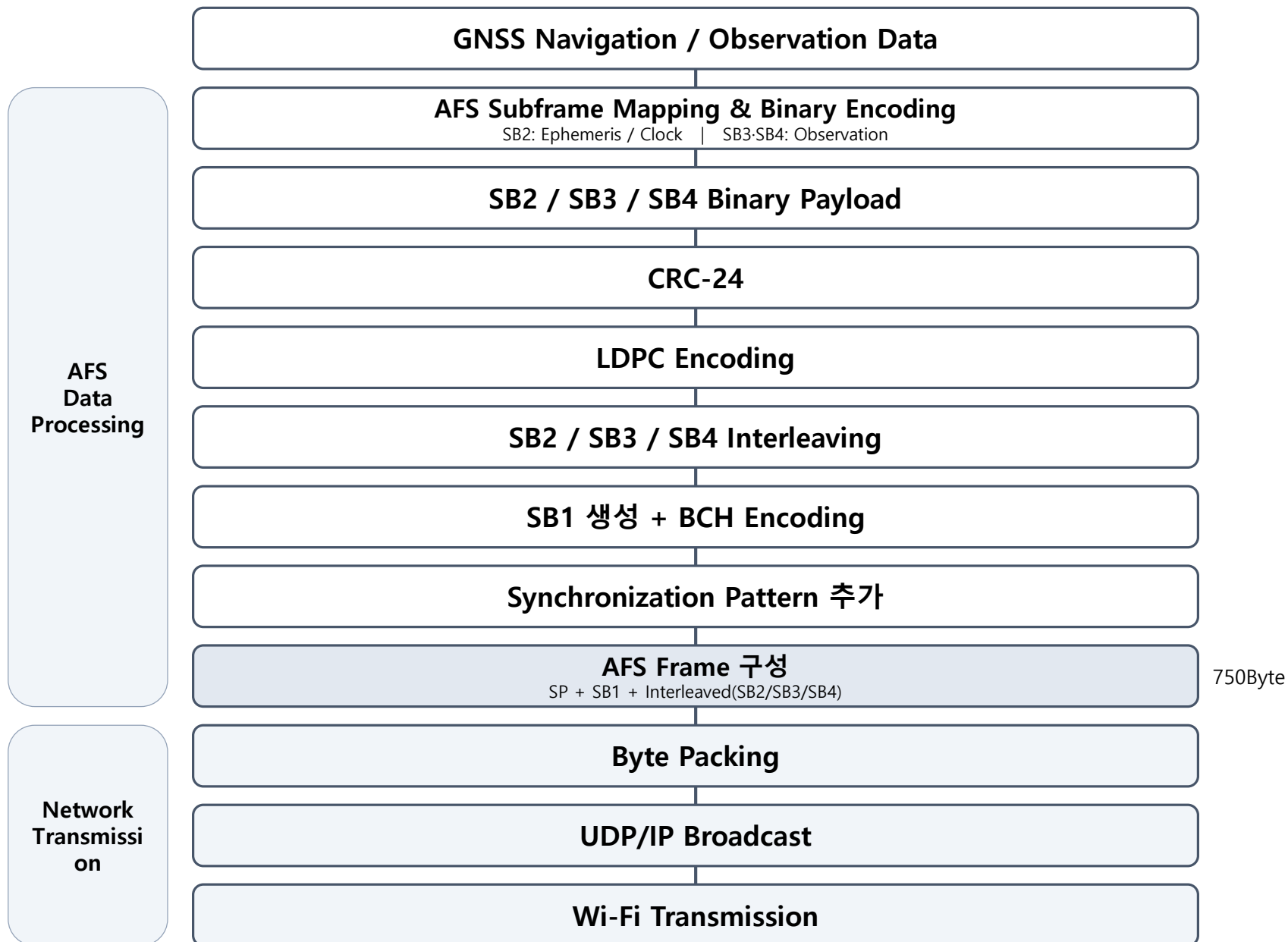
1. 시스템 구성도



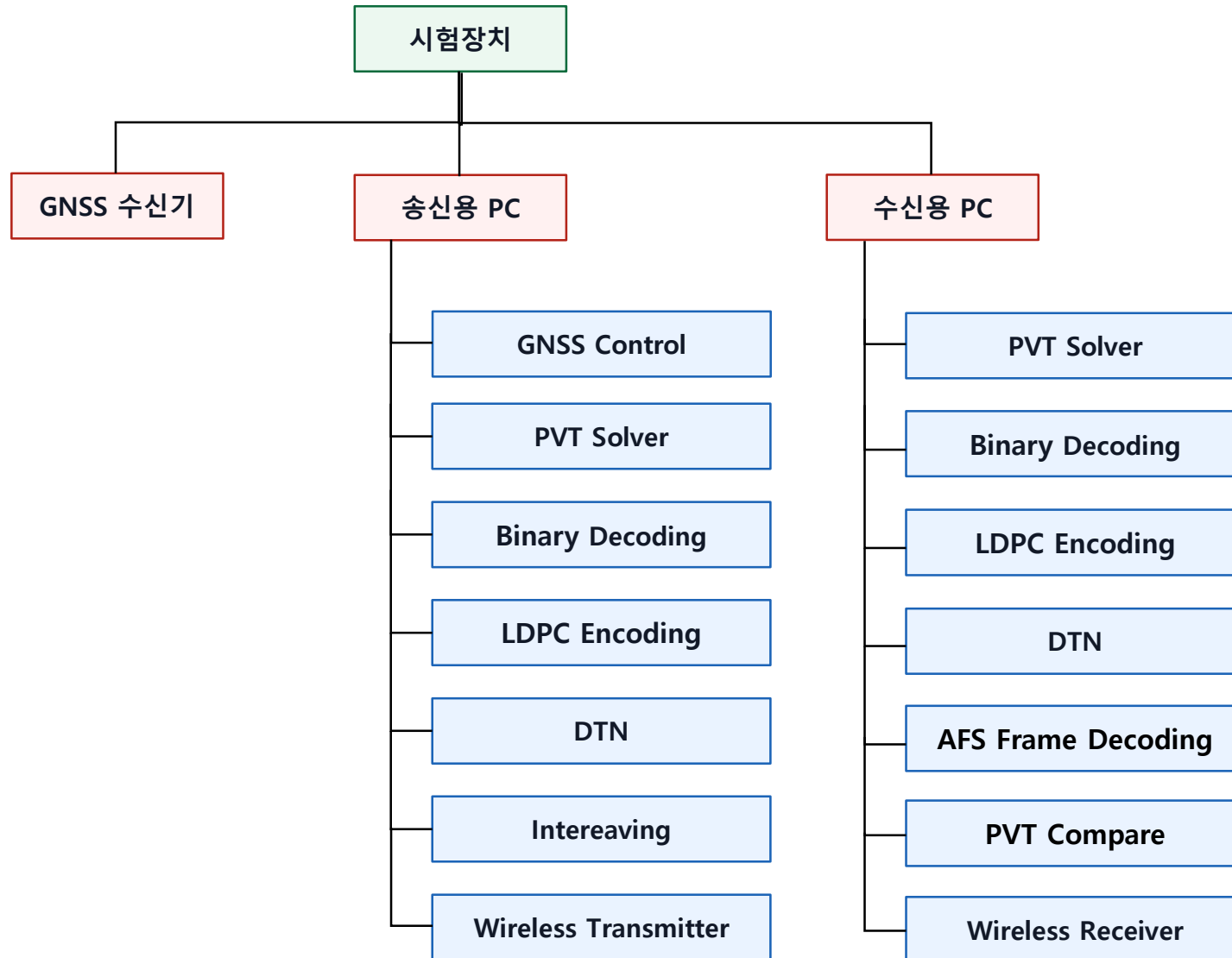
2. 전체 구성 블록도



3. 송신 데이터 흐름도



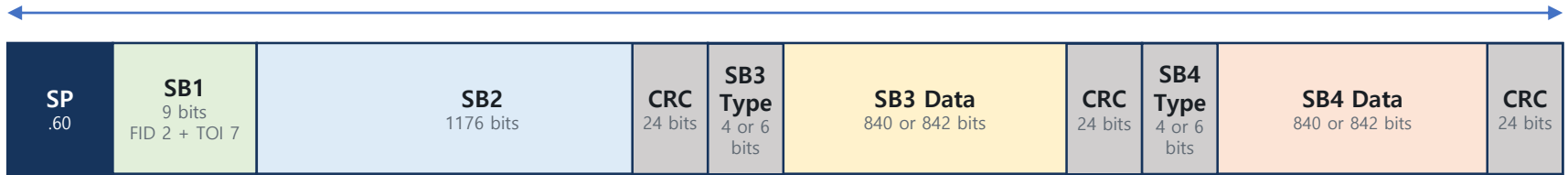
4. 시험장치 구성



5. AFS Frame 구조

AFS Frame

12sec



Fundamental Type

MSG-G4, G8, G2, G30

MSG-G5, S19, S20, G24, G32

MSG-G1

Optional Type

MSG-G3, G23, G10, G11, G14, G31, Custom

MSG-G3, G5, G10, G11, G14, G15, G17, G22, G23, G25, G27, G28, G29, G31, G32, Custom

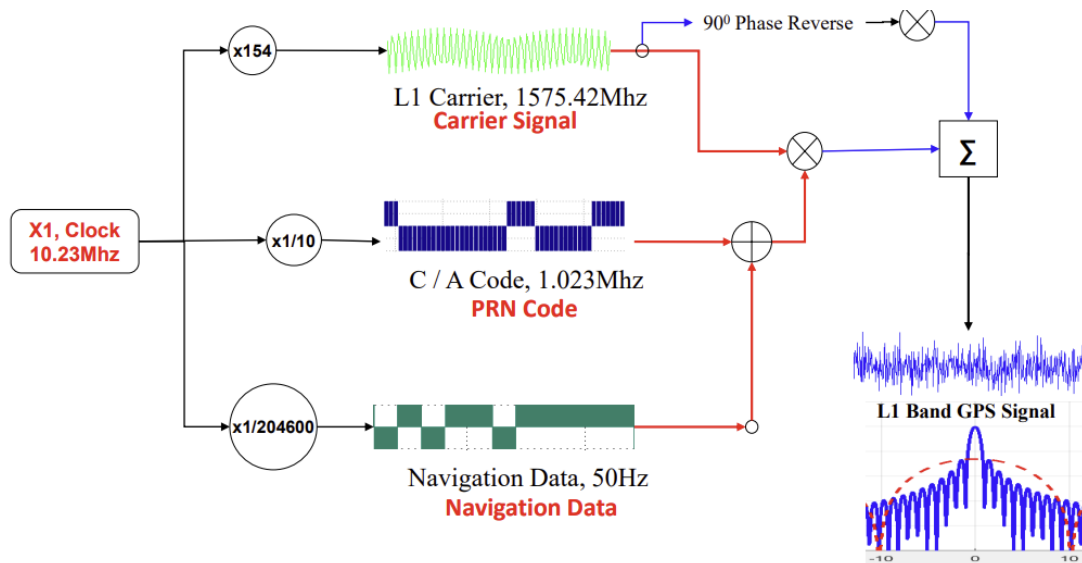
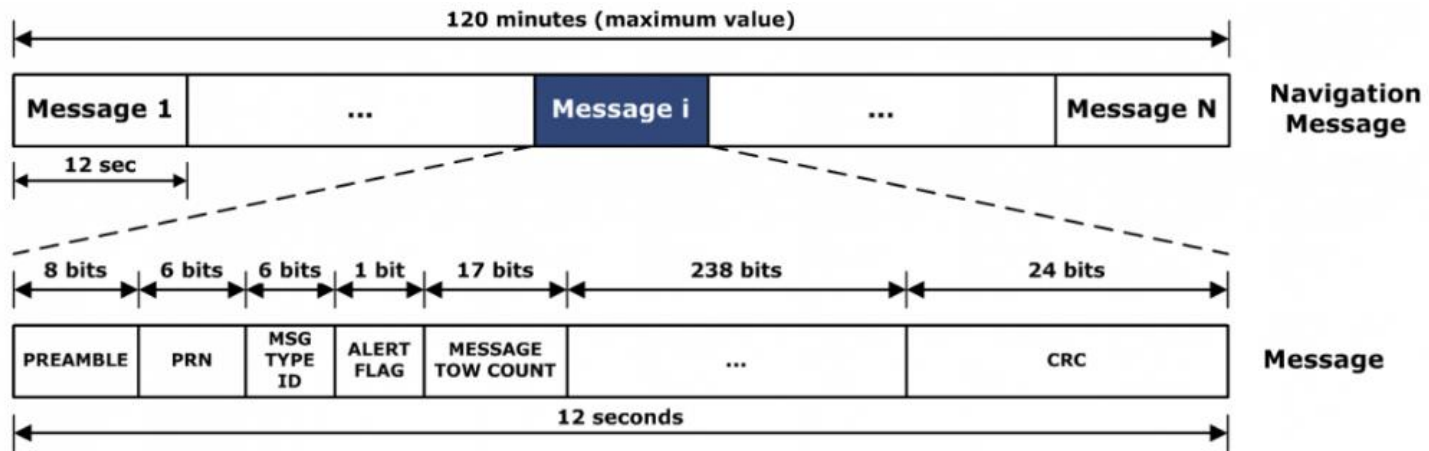
- SB3, SB4의 데이터는 프레임마다 반드시 포함 될 필요는 없음
- 데이터가 많으면 여러 프레임에 걸쳐 나눠 전송 가능
- Data Type에 대한 bit단위 내용은 정의 되어있지 않음
- 750Byte

6. LNIS Message → AFS Subframe Allocation

공식 메시지 배치와 본 시험 GNSS RAW Custom Message 위치

MSG ID	Message Title	Subframe	LNIS 1.0	시험 적용
G2	Health and Safety	SB2	Yes	참조
G4	Sorbit Ephemeris & Clock	SB2	Yes	AFS 필수정보
G5	MOrbit Almanac	SB3/SB4	Yes	선택
G8	Time of Transmission	SB2	Yes	AFS 시간
G12	Observation	SB3	Yes	미사용
Custom	LNSP-defined Custom Message	SB4	Allowed	GNSS RAW

7. GNSS 송신 데이터



8. RINEX Observation data(GNSS Raw Data)

2.11	OBSERVATION DATA					Mixed(MIXED)	RINEX VERSION / TYPE	
cnvtToRINEX 3.08.0	convertToRINEX OPR					13-Feb-19 17:58 UTC	PGM / RUN BY / DATE	
0034							COMMENT	
0034							MARKER NAME	
GNSS Observer	Trimble						MARKER NUMBER	
5736470034	TRIMBLE R10					5.30	OBSERVER / AGENCY	
	TRMR10					NONE	REC # / TYPE / VERS	
							ANT # / TYPE	
							APPROX POSITION XYZ	
							ANTENNA: DELTA H/E/N	
-8924.2185	-5507652.1952	3205740.9469						
1.5850	0.0000	0.0000						
1	1	0						
5	C1	C2	L1	L2	P2			
1.000								
2019	1	5	16	24	11.0000000			
2019	1	5	17	31	27.0000000			
0								
18								
22								
G10	4035	4022	4022	8044	4022			
G12	3986	3939	3953	7869	3930			
G14	4030	0	4028	4027	4028			
G20	4030	0	4010	4008	4008			
G21	2712	0	2706	2706	2706			
G24	1853	1842	1845	3674	1832			
G31	4034	4028	4028	8046	4018			
G32	4033	4028	4028	8056	4028			
CARRIER PHASE MEASUREMENTS: PHASE SHIFTS REMOVED							COMMENT	
19	1	5	16	24	11.0000000	0	END OF HEADER	
20673894.914	7	20673897.582	9	108642047.509	17	84656157.178	19	20673898.500
24514584.992	5	24514588.410	8	128825021.257	15	00383159.629	18	24514588.453
22096320.398	6			116116947.090	16	190480756.534	57	22096322.105
20649612.984	6			108514449.034	16	84556718.979	58	20649615.223
22727231.258	5			119432399.441	15	93064217.601	56	22727230.508
23548843.852	5	23548849.438	8	123750006.770	15	96428598.626	18	23548848.832
23770864.883	6	23770867.926	8	124916742.230	16	97337721.145	18	23770868.449
21280769.086	7	21280773.453	9	111831197.425	17	87141209.380	19	21280773.898
19	1	5	16	24	12.0000000	0		
20673787.609	7	20673790.762	9	108641484.816	7	84655718.718	9	20673791.293
24514314.883	5	24514318.414	8	128823601.798	5	100382053.557	8	24514318.426
22096122.836	6			116115910.326	6	90479948.674	47	22096124.977
20649828.898	6			108515582.444	6	84557602.163	48	20649830.723
22727841.766	5			119435610.858	5	93066720.019	46	22727840.875
23549497.172	5	23549502.707	8	123753438.215	5	96431272.471	8	23549502.316
23770183.133	6	23770185.832	8	124913161.207	6	97334930.740	8	23770186.344
21280692.422	7	21280696.422	9	111830793.490	7	87140894.628	9	21280696.699

Observation Data for Satellite G10, G12, ...

C1, C2, P2: Pseudo Range (Distance to Satellite)
L1, L2: Carrier Phase

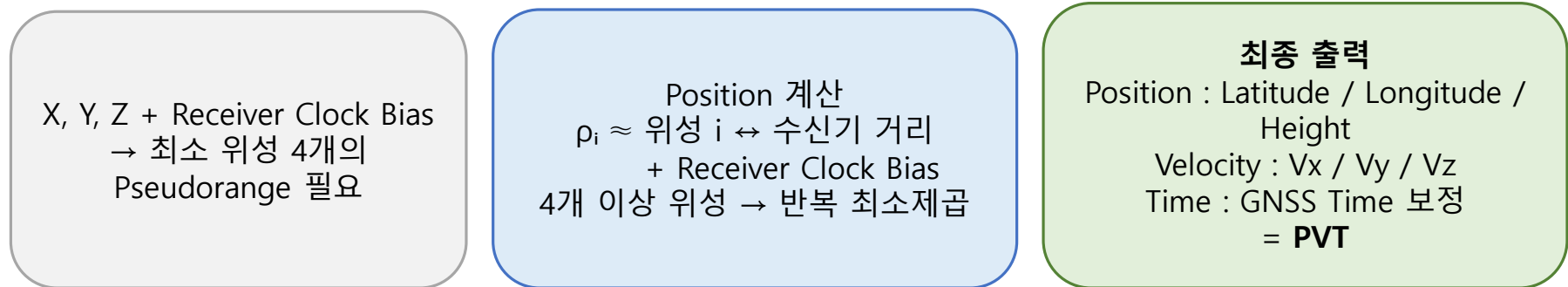
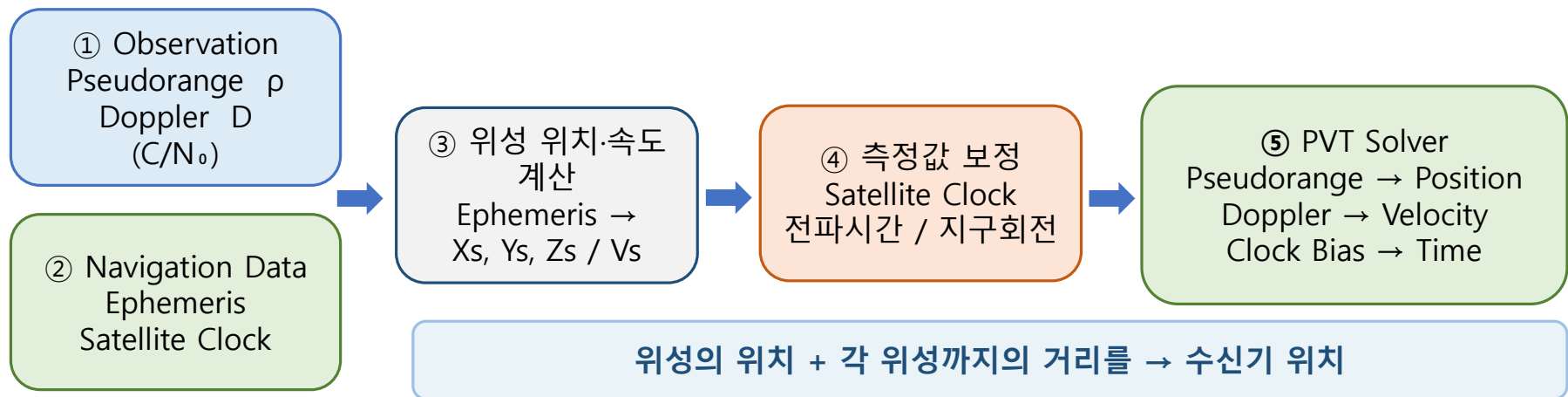
Yellow Field: Loss of Lock Indicator
 0 or blank: OK or not known
 Bit 0: Lost Lock
 Bit 1: Opposite Wavelength Factor
 Bit 2: Under Antispoofing

Behind Yellow Field: Signal Strength
 0 or blank = unknown, 1 = min, 9 = max

9. RINEX Navigation Message(GNSS Raw Data)

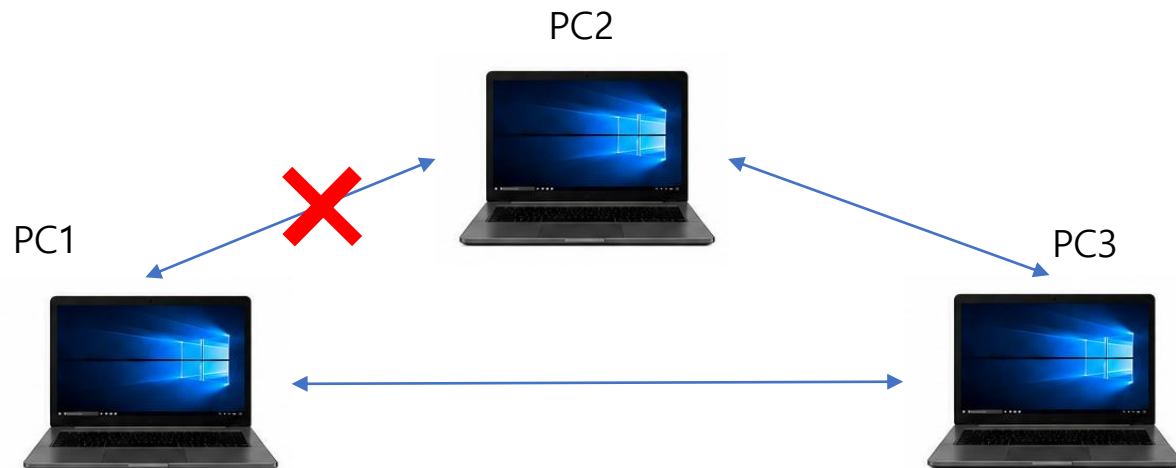
2.11	NAVIGATION DATA				GPS (GPS)	RINEX VERSION / TYPE	
cnvtToRINEX 2.90.0	convertToRINEX	OPR	05-Jul-17	03:38	UTC	PGM / RUN BY / DATE	
-----						COMMENT	
0.8382D-08	0.2235D-07	-0.5960D-07	-0.1192D-06			ION ALPHA	
0.8602D+05	0.6554D+05	-0.1311D+06	-0.4588D+06			ION BETA	
-0.931322574615D-09	-0.355271367880D-14	405504	1947			DELTA-UTC: A0,A1,T,W	
18						LEAP SECONDS	
						END OF HEADER	
Navigation Data							
PRN 32	17	05	01	00	00	0.0-0.400723423809D-03-0.110276232590D-10	0.000000000000D+00
						0.370000000000D+02-0.806250000000D+01	0.455840416154D-08-0.192420920137D+01
						-0.353902578354D-06	0.111064908560D-02
						0.826455652714D-05	0.515371503258D+04
						0.864000000000D+05-0.782310962677D-07	0.675647076441D-01-0.838190317154D-07
						0.958529124300D+00	0.221156250000D+03-0.265074890978D+01-0.796390315710D-08
						-0.389659088008D-09	0.100000000000D+01
						0.194700000000D+04	0.000000000000D+00
						0.240000000000D+01	0.000000000000D+00
						0.465661287308D-09	0.370000000000D+02
						0.795120000000D+05	0.400000000000D+01
						0.000000000000D+00	0.000000000000D+00
PRN 24	17	05	01	00	00	0.0-0.341213308275D-04-0.454747350886D-12	0.000000000000D+00
						0.100000000000D+02	0.787812500000D+02
						0.459340561950D-08	0.167267059468D+01
						0.404566526413D-05	0.564297637902D-02
						0.102464109659D-04	0.515370226479D+04
						0.864000000000D+05-0.782310962677D-07	0.108986675687D+01
						0.484287738800D-07	0.945651423640D+00
						0.170906250000D+03	0.490563049326D+00-0.815641117584D-08
						-0.128933942045D-09	0.100000000000D+01
						0.194700000000D+04	0.000000000000D+00
						0.240000000000D+01	0.000000000000D+00
						0.279396772385D-08	0.100000000000D+02
						0.792180000000D+05	0.400000000000D+01
						0.000000000000D+00	0.000000000000D+00

10. PVT Solve



RINEX Observation + RINEX Navigation → PVT

11. HDTN 프로토콜 시험구성



순서	DTN 시험
①	PC1 → PC2 정상 Bundle 전달 확인
②	PC2에서 PC1의 DTN TCPCL Port만 차단
③	PC1-PC2 Contact 실패 확인
④	PC1이 Bundle을 Storage에 보관
⑤	대체 Contact PC1 → PC3 선택
⑥	PC3이 Bundle 저장/Forward
⑦	PC2 → PC3 전달
⑧	PC2에서 원본 Bundle 수신 확인